Instructions:

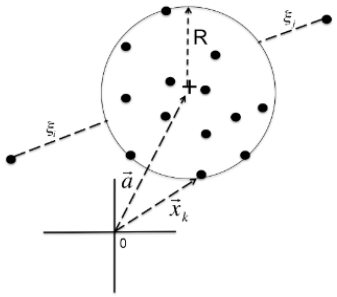
You will work on these exercises during the practical sessions in the fifth week of the course.

Below is an indication of the exercises that we will cover (at least) in each session. We encourage you try them before the session.

- First session (Wednesday): exercises 1 and 3.
- Second session (Friday/Monday): Multiple choice and True/False questions from last year's exams (available on Canvas).

Exercise 1: Kernel Outlier Detection

Consider the picture below. The dots represent data items. Our task is to derive an algorithm that detects the outliers. To that end, we draw a circle centered at location $\boldsymbol{a}$ with radius R. All data cases that fall outside the circle are detected as outliers.



We write down the primal program that finds such a circle:

$$\min_{\boldsymbol{a},R,\boldsymbol{\xi}} R^2 + C \sum_{i=1}^{N} \xi_i$$

subject to

$$\|\boldsymbol{x}_i - \boldsymbol{a}\|^2 \leq R^2 + \xi_i, \qquad \xi_i \geq 0, \qquad i = 1, \ldots, N.$$

In words, we want to minimize the radius of the circle subject to the constraint that most data cases should lie inside it. Outliers are allowed to stay outside, but they pay a price proportional to their distance from the circle boundary and C.

(a) Introduce Lagrange multipliers for the constraints and write down the primal Lagrangian. Use $\{\alpha_i\}$ for the first constraint and $\{\mu_i\}$ for the non-negativity constraint $\xi_i \geq 0$.

(b) Write down all stationary and KKT conditions. Hint: take the derivative with respect to R^2 instead of R.

(c) Use these conditions to derive which data cases $\boldsymbol{x}_i$ will have $\alpha_i > 0$ and which ones will have $\mu_i > 0$.

(d) Derive the dual Lagrangian and specify the dual optimization problem. Kernelize the problem, that is, write the dual program only in terms of kernel entries and Lagrange multipliers.

(e) The dual program will return optimal values for $\{\alpha_i\}$. Assume that at least one of these satisfies $0 < \alpha_i < C$. In terms of the optimal values for α_i, compute the optimal values for the other dual variables $\{\mu_i\}$. Then solve the primal variables $\{\boldsymbol{a}, R, \boldsymbol{\xi}\}$ in terms of the dual variables.

(f) Assume we have solved the dual program. We now want to apply it to new test cases. Describe a test in the dual space, in terms of kernels and Lagrange multipliers, that could be used to detect outliers. Students who got stuck along the way may describe the test in primal space.

(g) What kind of solution do you expect if we use $C = 0$? What solution do you expect if we use $C = \infty$?

Exercise 2: L2 Support Vector Machine

Exam question

Let $\{(\mathbf{x}_n, t_n)\}_{n=1}^N$ be a set of N training pairs of feature vectors $\mathbf{x}_n$ and labels $t_n \in \{-1, +1\}$. The following is the primal formulation of the L2-SVM, a variant of the standard SVM obtained by squaring the penalty for the slack variables:

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\mathbf{w}^\top\mathbf{w} + \frac{C}{2}\sum_{n=1}^N \xi_n^2 \\ \text{subject to} \quad & t_n(\mathbf{w}^\top\mathbf{x}_n + b) \geq 1 - \xi_n, \qquad n = 1, \ldots, N, \\ & \xi_n \geq 0, \qquad n = 1, \ldots, N. \end{aligned}$$

Note: the first subquestion is difficult, but it is not needed to solve the rest of the subquestions.

(a) Show that removing the constraints $\xi_n \geq 0$ does not change the optimal solution to the primal problem (i.e., proof that the constraints $\xi_n \geq 0$ are no longer necessary in this L2-SVM formulation).

Hint: assume there is an optimal solution such that $\xi_n < 0$ for some j and try reach a contradiction.

After removing the non-negativity constraints for the slack variables, we obtain the relaxed optimization problem used in the rest of the exercise:

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \frac{1}{2}\mathbf{w}^\top\mathbf{w} + \frac{C}{2}\sum_{n=1}^N \xi_n^2 \\ \text{subject to} \quad & t_n(\mathbf{w}^\top\mathbf{x}_n + b) \geq 1 - \xi_n, \qquad n = 1, \ldots, N. \end{aligned}$$

(b) Write the Lagrangian.

(c) List the KKT conditions. How many scalar conditions are there?

(d) Derive the stationary conditions for the Lagrangian.

(e) Derive the dual Lagrangian and state the dual optimization problem.

(f) We obtain an optimal solution $\boldsymbol{\alpha}^*$ for the dual problem. Compute the optimal primal variables $\mathbf{w}^*$, b^*, and $\boldsymbol{\xi}^*$ from that value.

(g) How can we use the optimal parameters $(\mathbf{w}^*, b^*)$ to classify a test datapoint $\mathbf{x}$?

Exercise 3: Gaussian Processes

Exam question

(a) What is a Gaussian Process? Is it a supervised or an unsupervised method?

(b) In a GP, we treat the function value $f(x)$ at a specific position as a random variable. If we have two positions x and x', how is the correlation between the random variables $f(x)$ and $f(x')$ defined in a Gaussian Process?

(c) Consider the standard Gaussian Radial Basis Function (RBF) or Squared Exponential kernel: $k(x, x') = \sigma^2 \exp\left(-\frac{\|x-x'\|^2}{2\ell^2}\right)$.

- What does this kernel imply about the function properties?
- Is there a parameter that controls how "smooth" or "wiggly" the sampled functions are? How would you change it to increase smoothness/correlation?

(d) In other ML methods like Support Vector Machines (SVMs), we also use kernels (the "kernel trick"). Why do we do this?